

SMART-HORIZON — Figure Reference

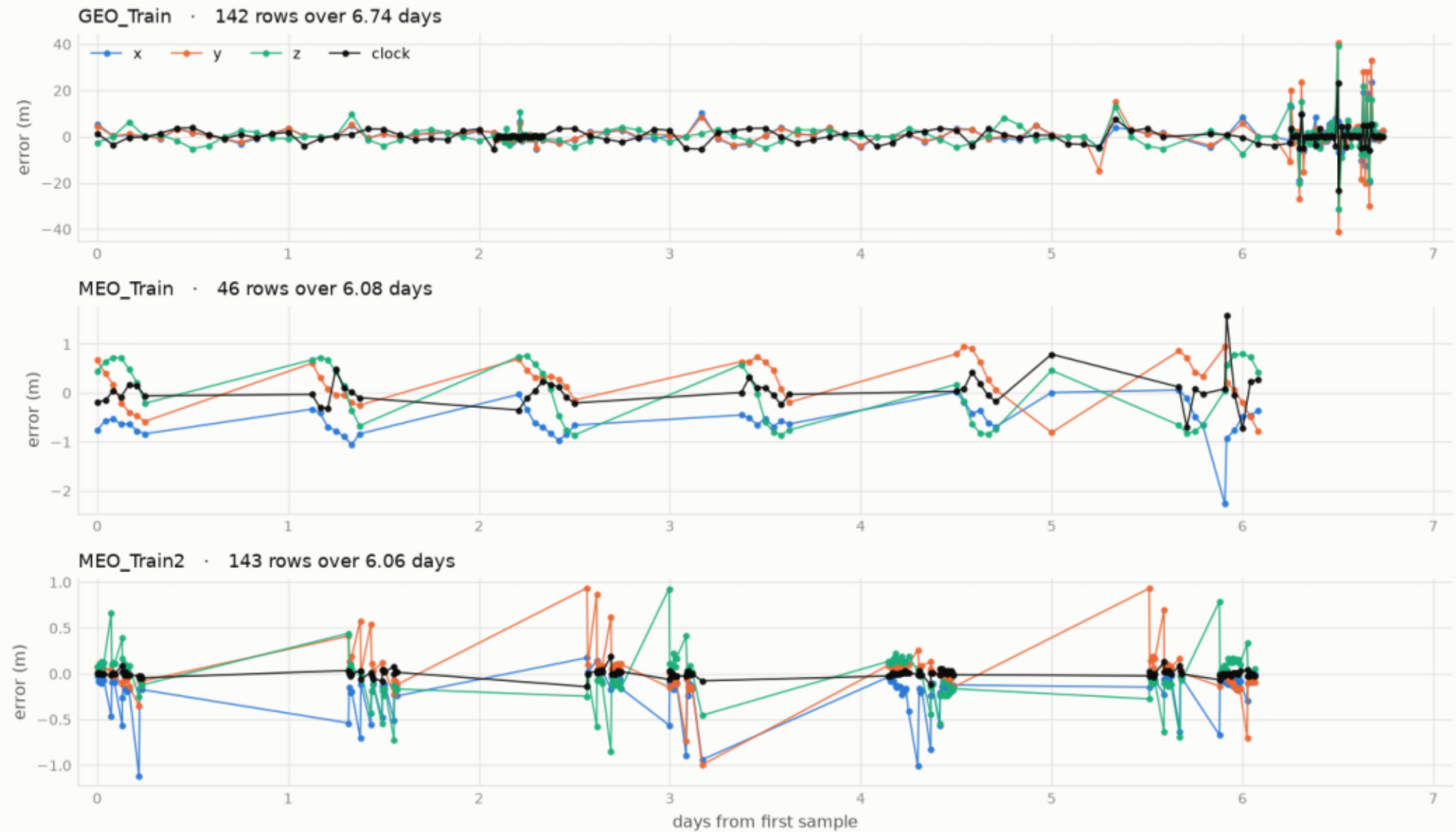
Every image referenced in the demo script, in order,
labelled by Act (①–⑧) and source filename (Stage-04/figures/).

For the teammate integrating the deck: match the Act number to DEMO_SCRIPT.md.

The three training files — X/Y/Z/clock error, GEO vs MEO scale.

The three training files, exactly as delivered

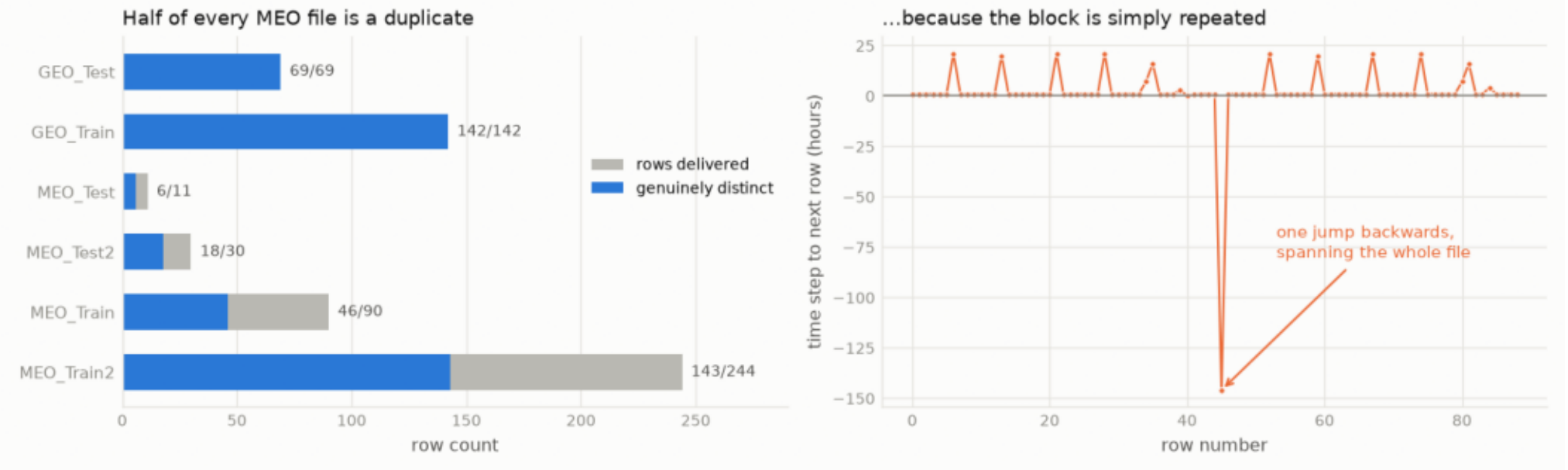
Four error channels each. Note the y-axis: the GEO file (top) runs to ± 40 m, the MEO files to about ± 1 m.



~Half of every MEO file is duplicated rows (one test set = 6 pts).

The duplication is a copy-paste, so removing it is safe

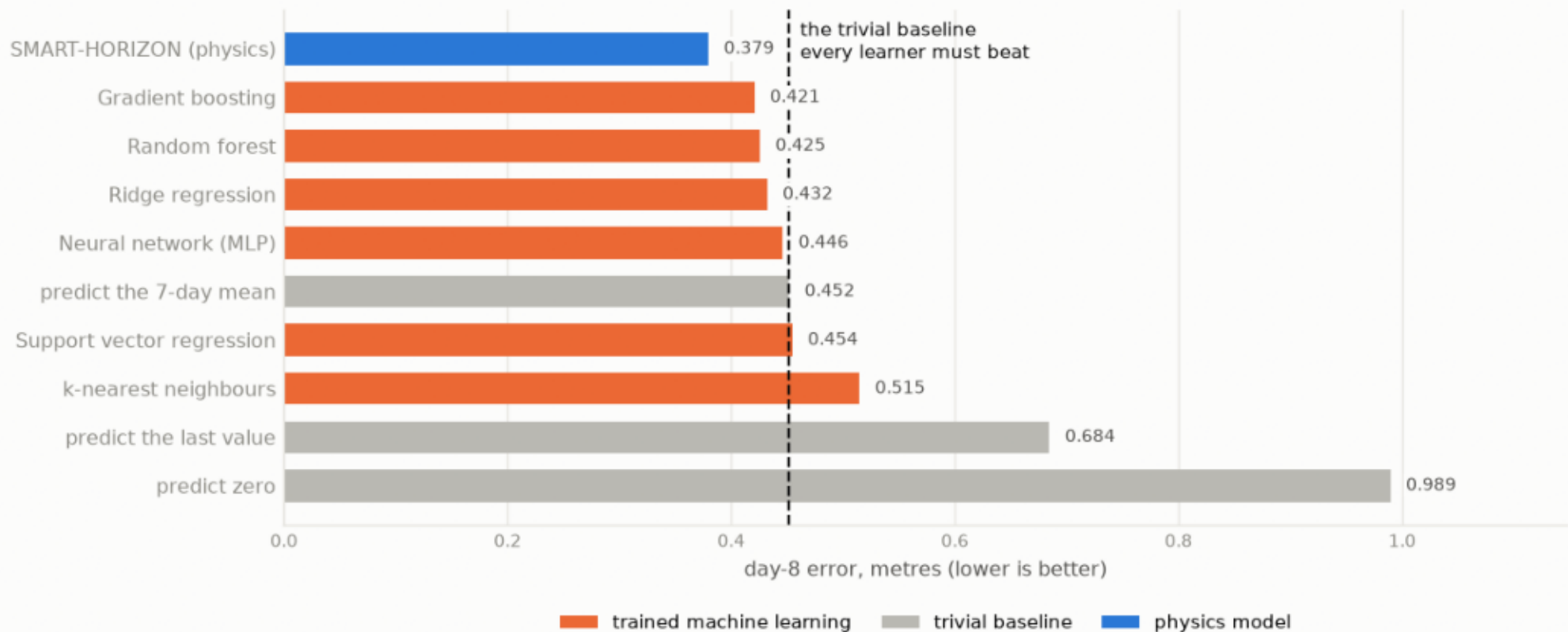
Exactly one backwards time step per file, spanning its full length. Nothing is lost by de-duplicating.



Six standard ML models all plateau at the trivial baseline.

Six standard ML models, trained properly — and they stall at the baseline

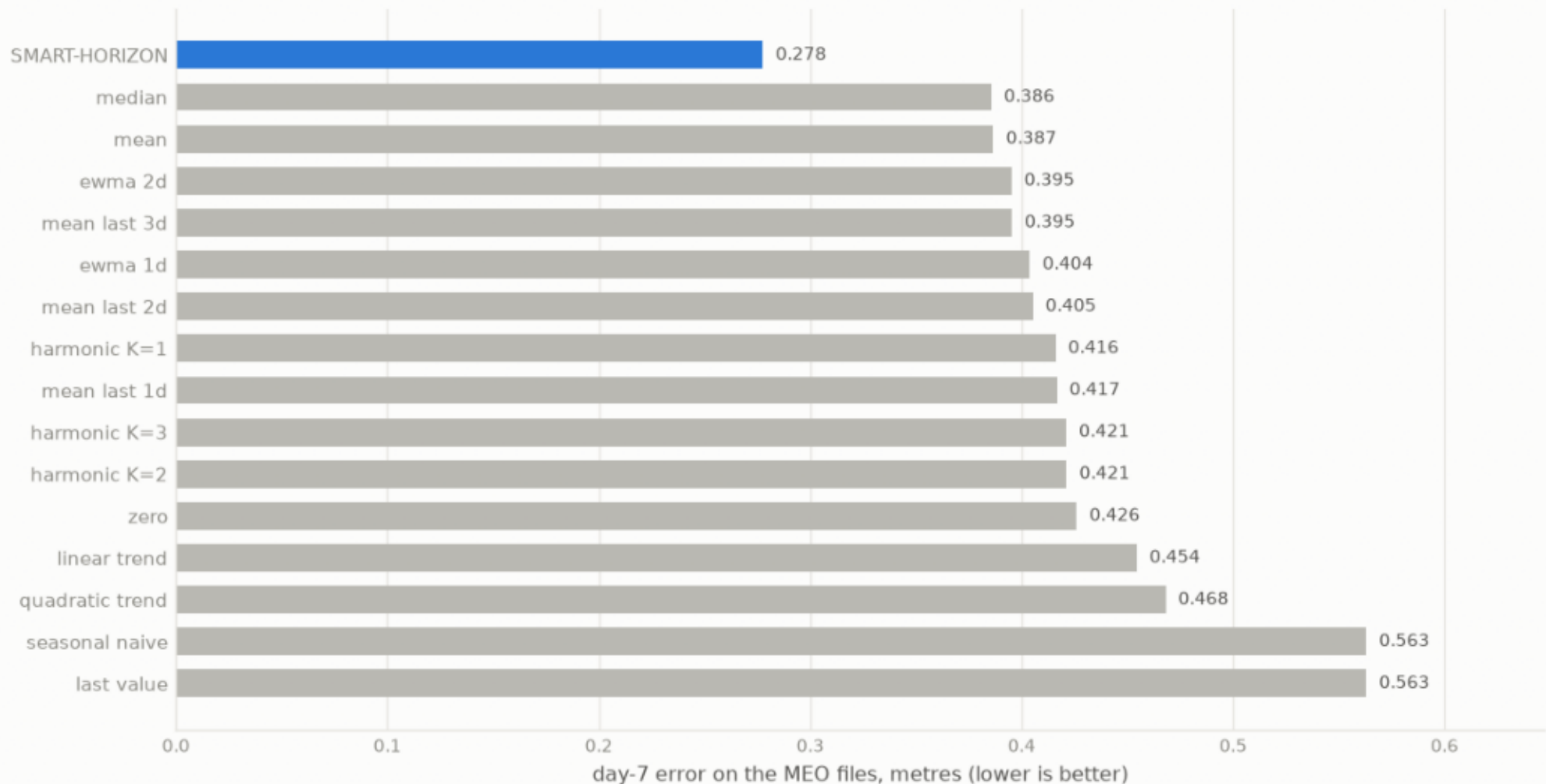
400 windows, 27,000 training rows, 19 engineered features. The learners are not weak; the five columns simply do not contain what decides day 8.



16 methods ranked — everything caps in the same narrow band.

Sixteen methods, ranked using only the delivered data

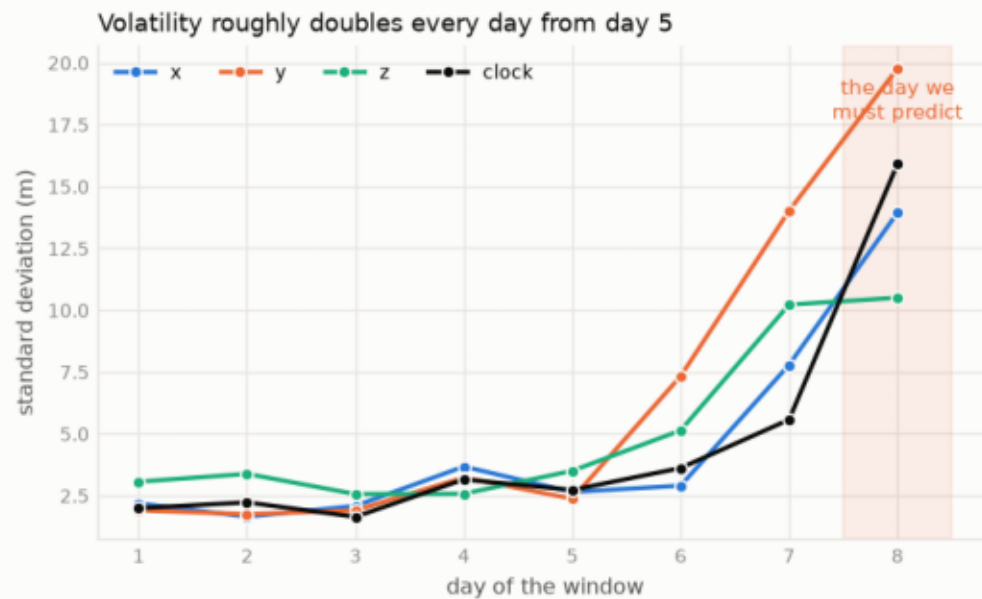
Protocol: fit on days 1-6 of their own file, score on day 7. No outside data.



GEO volatility ramps day-by-day; day 8 continues the climb.

The GEO file gets steadily wilder through the week

This is a ramp, not a glitch. Any model fitted on the quiet early days is being scored on the wild end.



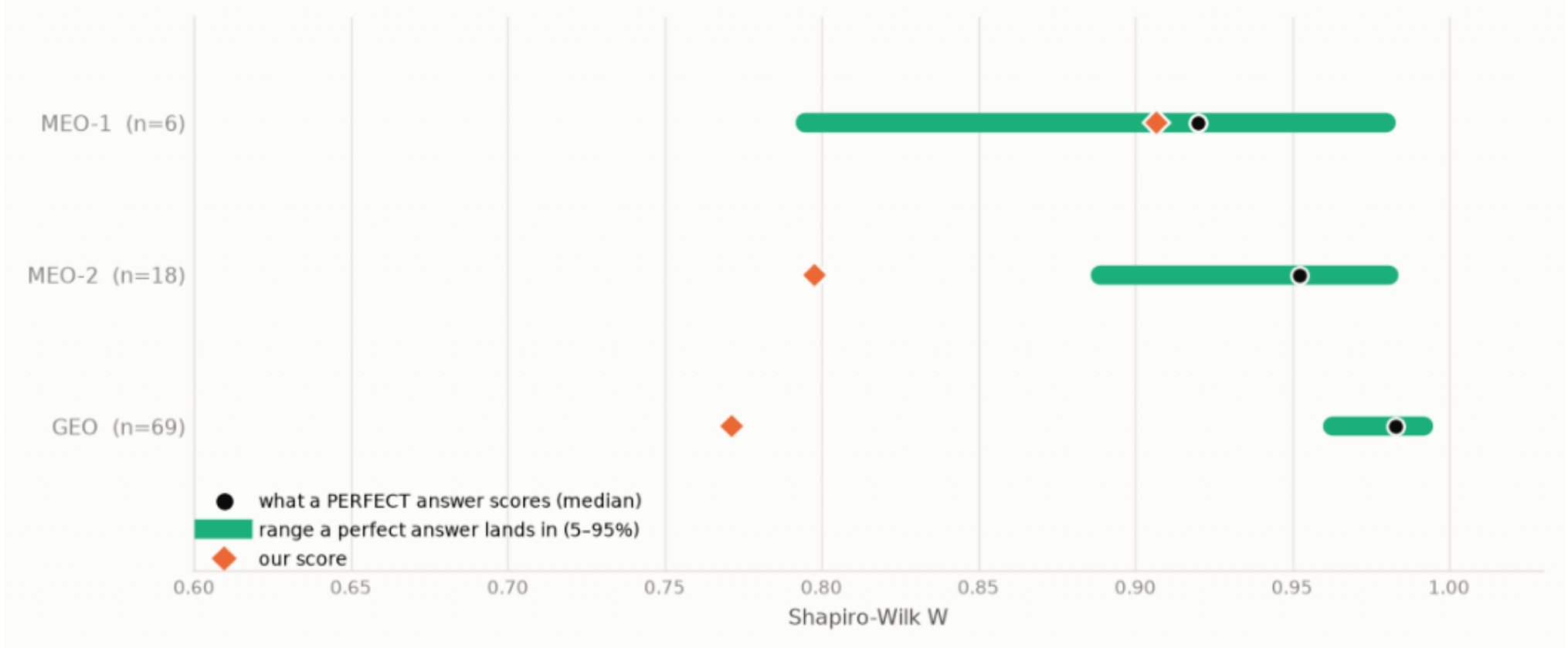
The same thing, seen directly



Even a perfect model can't score ~ 1.0 at $n=6/18/69$ (lottery).

A flawless answer cannot score 1.0 — and at $n=6$ it is mostly luck

Green is where a model with perfectly random errors lands. With 6 test points that is anywhere from 0.79 to 0.98.

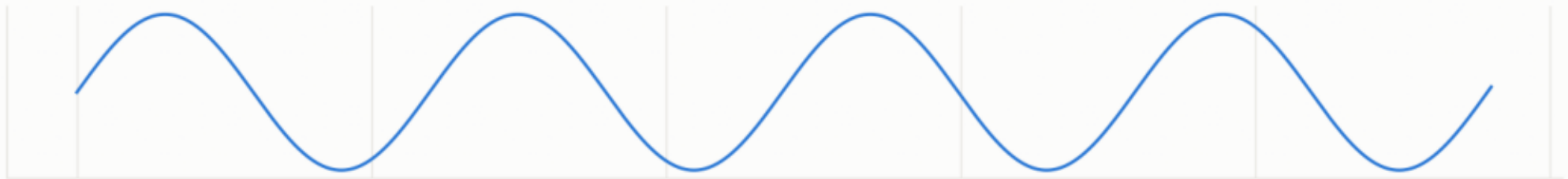


ECEF frame splits X/Y into two beat periods; Z keeps the orbital period.

Why the x and y axes hide two rhythms, not one

$\sin(A) \cdot \cos(B) = \frac{1}{2}[\sin(A-B) + \sin(A+B)]$. The z axis is the rotation axis, so it is untouched and stays at T.

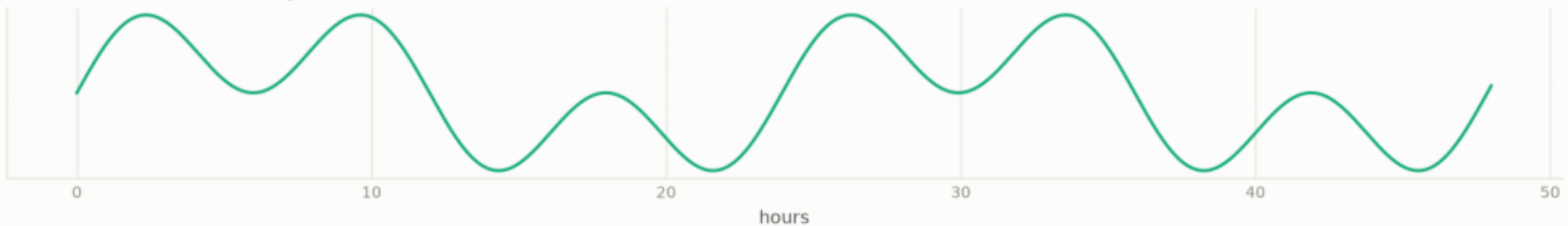
1. The error rotates with the satellite · period $T = 11.97$ h



2. The coordinate frame rotates with the Earth · period $T_{\text{sid}} = 23.93$ h



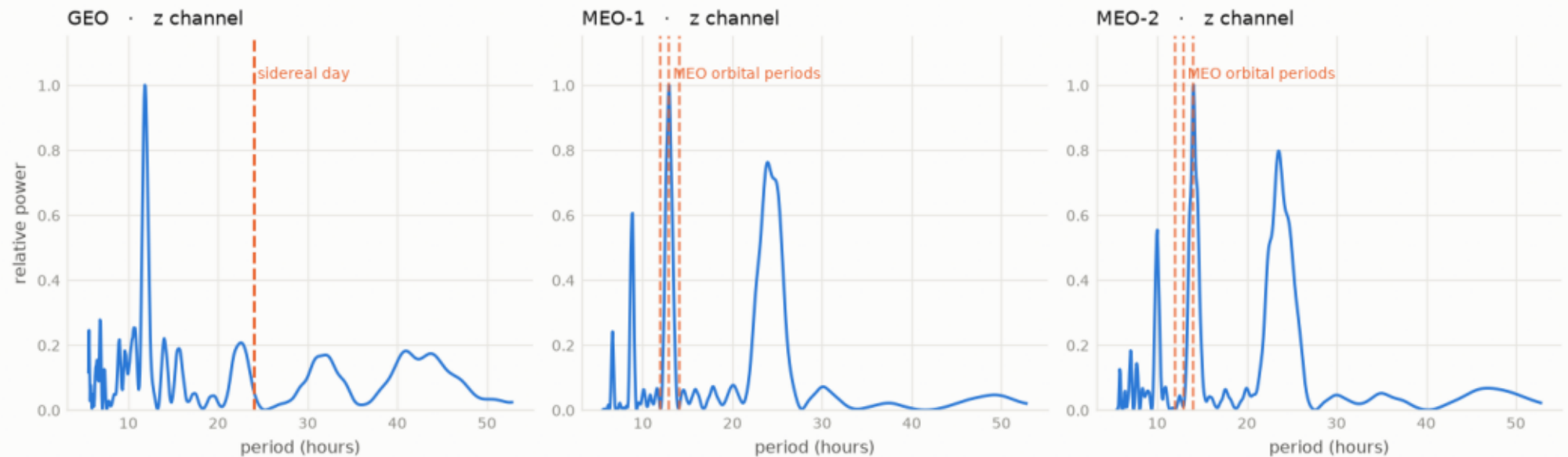
3. What we observe is their product — two new tones at 7.98 h and 23.93 h



Physics-predicted periods (marked before looking) land on the peaks.

The rhythms physics predicts are present in their own data

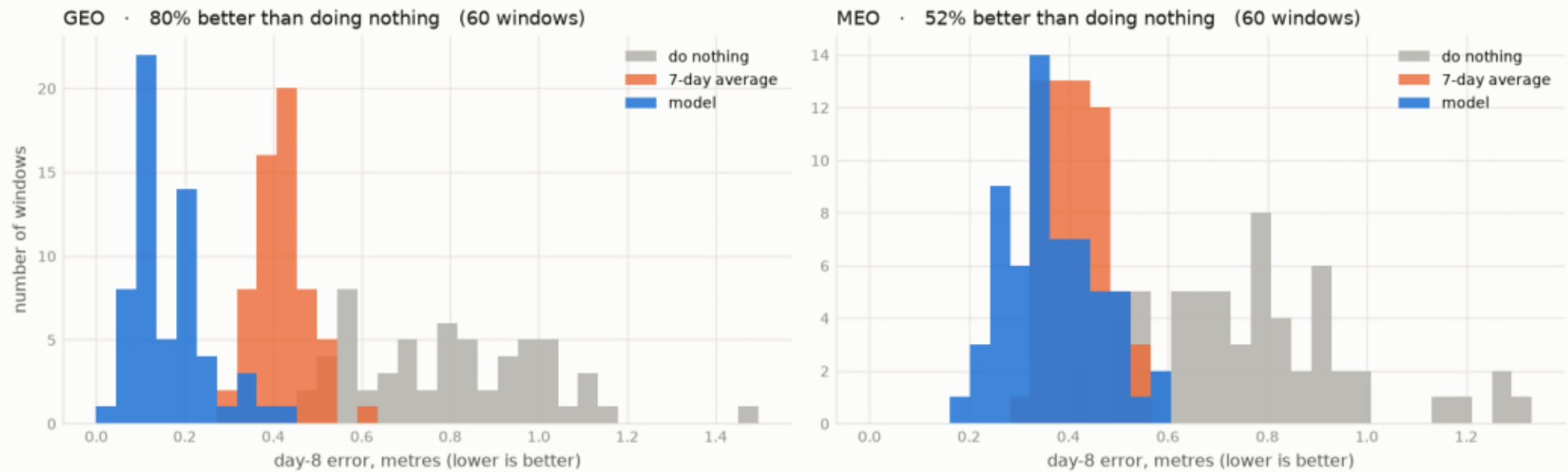
Orange dashed lines are placed by orbital mechanics alone, before looking at the data.



120 synthetic windows: model recovers day 8, beats do-nothing.

Not one lucky window — 60 of them, per orbit class

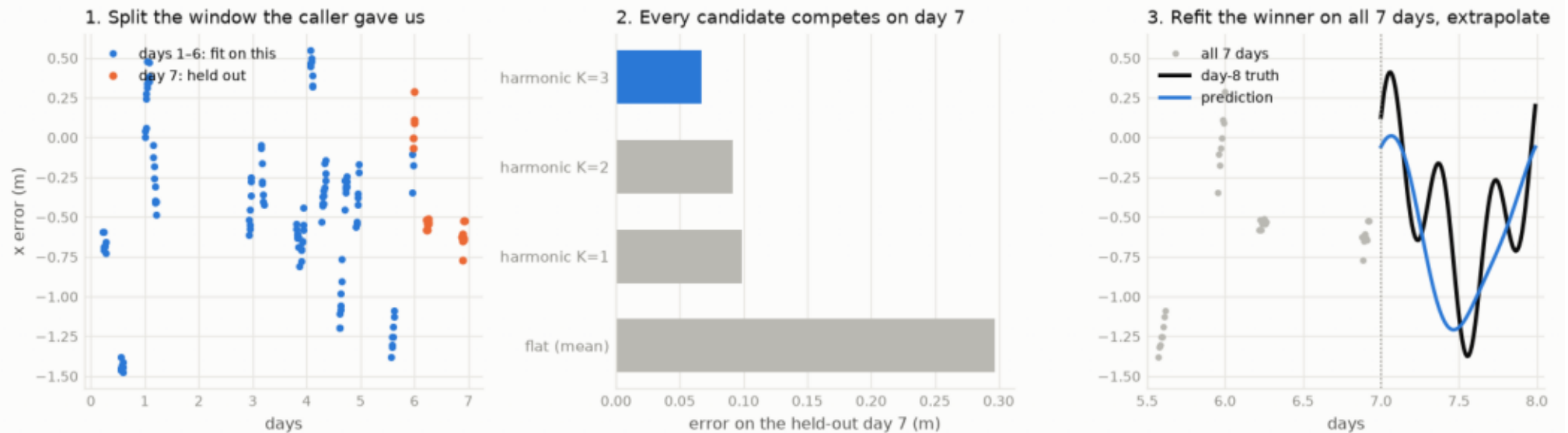
Each window is an independent signal with a different period, phase, drift and noise draw.



The model: physics fixes rhythms -> window self-selects basis -> ridge fits.

How the model works — the window grades itself

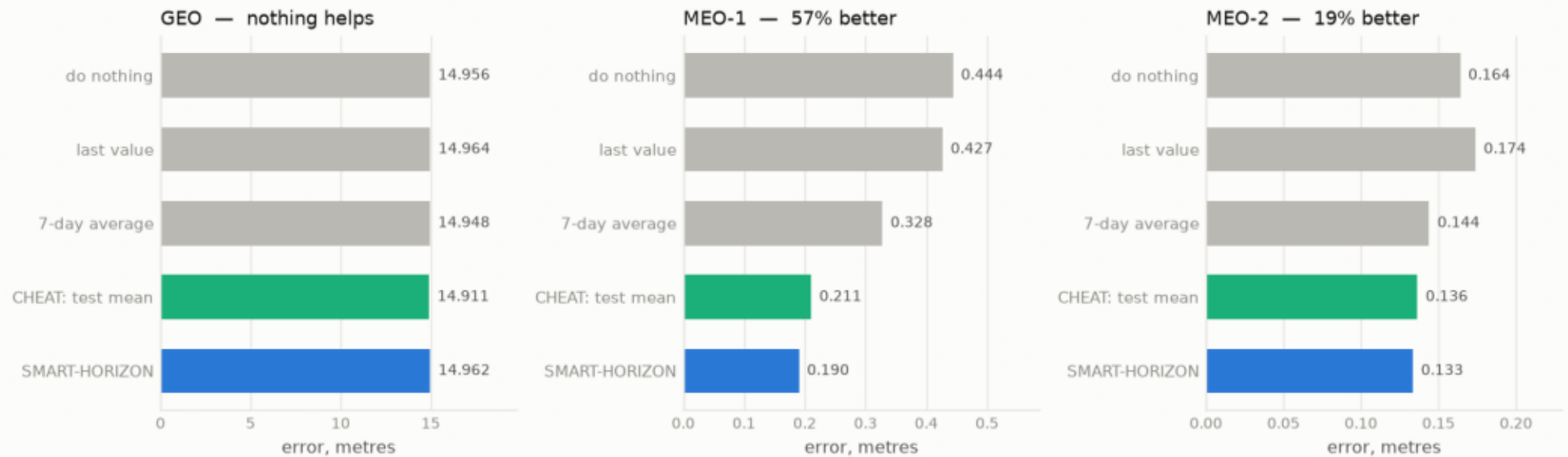
No stored training set is consulted at any point. The only model-selection signal is the caller's own day 7.



On real MEO: beats do-nothing AND the answer-key predictor.

On MEO the model wins. On GEO every method scores the same.

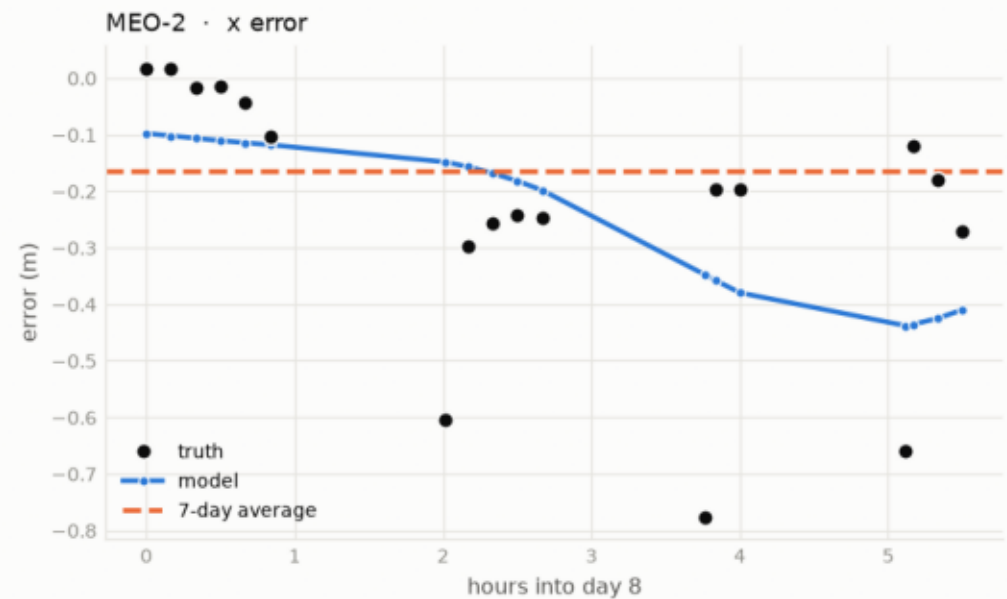
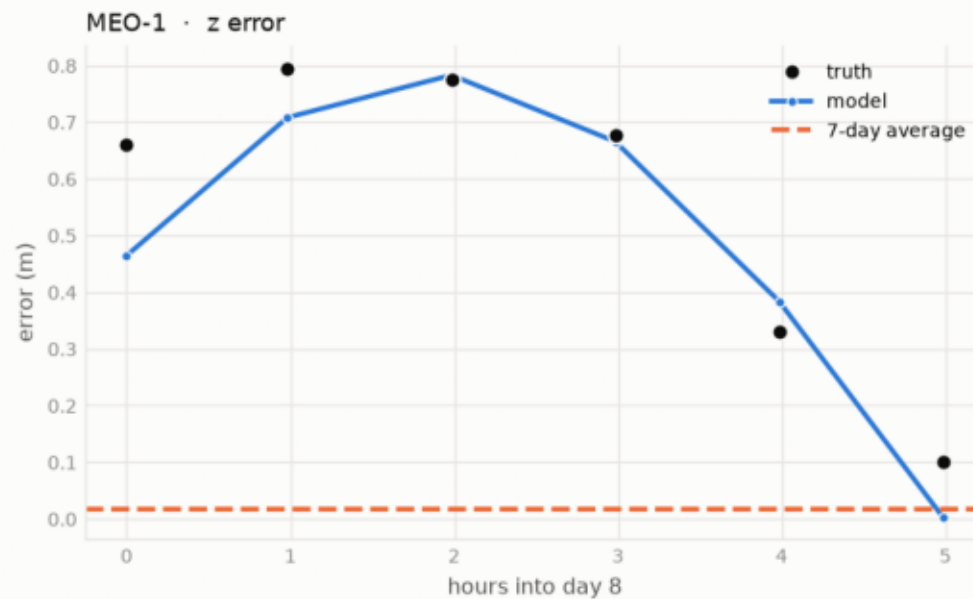
'CHEAT' is shown the answer key's own average — no legitimate model can use it. On MEO we beat it anyway.



(optional add-in) model traces the daily arc; average sits flat.

The model follows the shape of day 8, not just its level

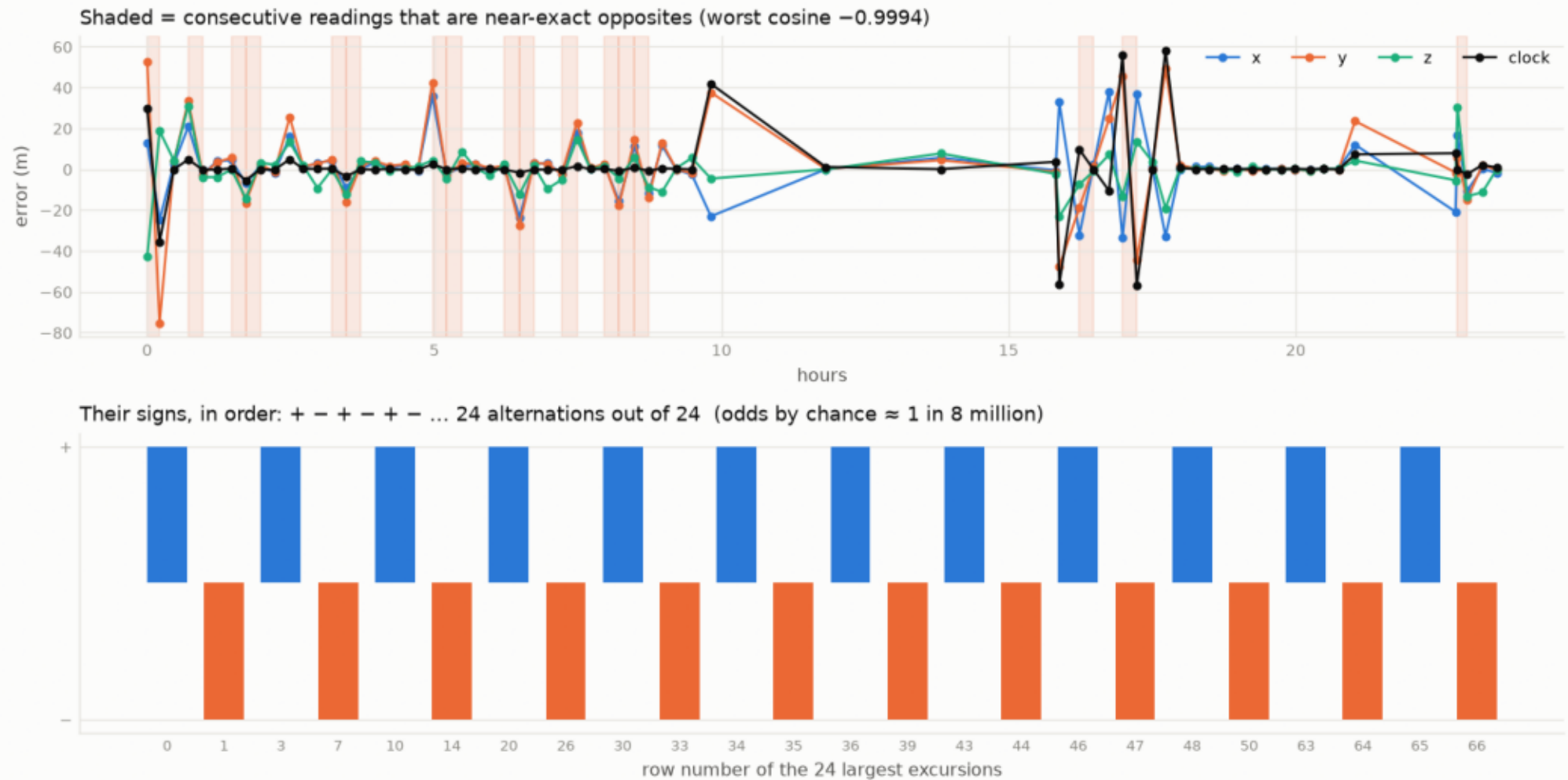
Left is a clean case, right a noisy one. Both are shown; neither is selected.



GEO file is non-physical: consecutive readings near-exact opposites.

The GEO file alternates sign with machine regularity

All four channels flip together. Position and clock come from unrelated physical mechanisms and cannot do that.



Inject the artefact into a clean signal -> same collapse (repro).

Proof by construction: the artefact alone is what breaks GEO

Identical underlying physics in both rows. The only change is the alternating sign pattern copied from their GEO file.

